

Confidential

Design Change
Request No. 25_128

Customer project Name/ 客户项目名称: JIM-4JJ MCR No. / MCR号:

Sample status / 样件状态: ☐ A Sample ☐ B Sample ☐ C Sample ☒ D Sample

Change part&product Name/更改零部件产品的名称(简要描述)

DOC DPF :F01ZH003EA-00
SCR: F01ZH003EB-00
DPF BRACKET:F01Z5017P5-05

Reason of changes/更改理由 (简要描述):

DPF 出气法兰, 客户装上双头螺柱后, 有密封垫片, 安装不上的风险, 需要增加 对出气法兰螺母的位置检测
SCR 客户要求 在尾管处增加纸质标签
侧支架 客户要求 更改客户料号

Change from/变更来源:

Customer request / 客户要求 ☒ Supplier request / 供应商请求 ☐ Design optimization / 设计优化 ☐
Correction / 更正(错误) ☐ Other / 其他 ☐ ()

Change inform to/变更通知人: Zhang Shengchang,Zhang Hanquan,Shen Weibo, Li Yicen, Fang Jiayan, Yin Kangzhuang
Yang Guangtuo,Jia Kebin, Luo jing, Lu Qingsong,Xie Youfu

Whether this change impact others component or product 此变更是否影响到其他部件和产品: ☐ yes/是 ☒ no/否

If impact others product, Whether others projects&product change or not? 如果影响, 别的部件和产品是否也需要同时变更?

Design / 设计工程师

Signature / 签名:

马坤阳

Date / 日期:

2025.12.01

Check / 校对工程师

Signature / 签名:

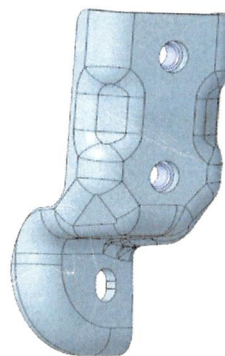
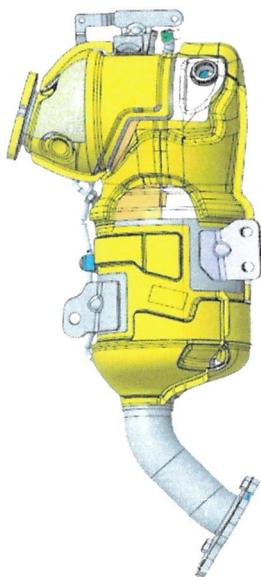
杨广拓

Date / 日期:

2025.12.02

Description of Change/更改描述: (下面空白处请列出详细的变更前后对比)

基于C样件模型未更改
详细更改点见附件



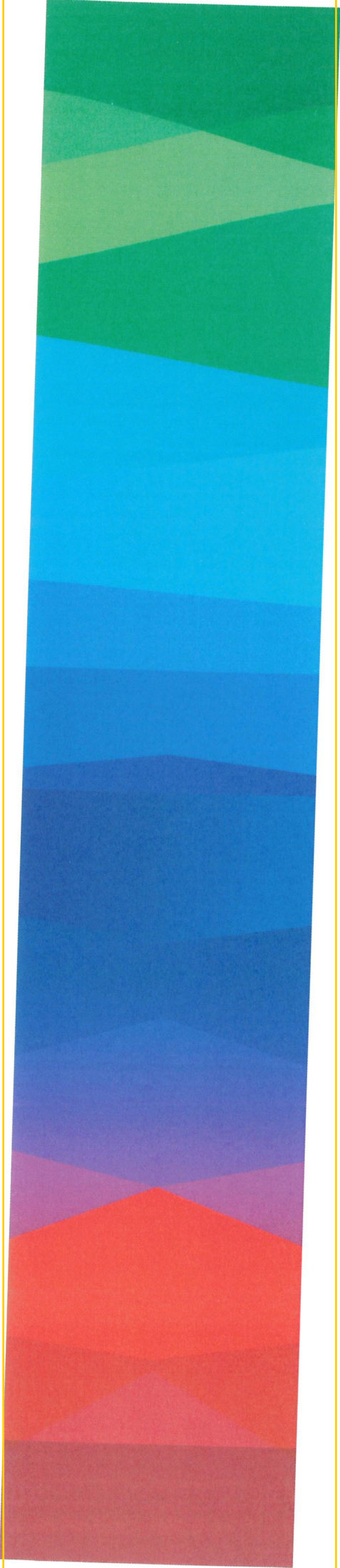
 Confidential	Design Change Review Sheet 设计变更评审表	Version 13 Effective date: 2025-09-25th
1. Function&Performance will be influenced? / 系统性能影响? ☐ yes/是 ☒ no/否		Design Change Request No.
If Yes, System impact evaluation? 如果是, 系统影响评估? (如果是初次样件放行, 空白处请说明样件目的) Whether need sample to test? Plan? 是否需要样件进行验证? 计划? Whether the catalyst information is confirmed? 催化剂信息是否确认? (物料号、图纸、SPEC)		System/签名 <u>沈伟波</u>
2. Interface&boundary (internal and customer) will be influenced? / 接口&边界(内部和客户)影响? ☐ yes/是 ☒ no/否		Catalyst/签名 <u>张汉泉</u>
If impact customer ,whether get confirm from customer side? 请说明影响和是否得到客户的确认?		Design /签名 <u>马坤阳</u>
3. Mechanical strength&Durability will be influenced? / 机械强度&可靠性影响? ☐ yes/是 ☒ no/否		
If Yes, How to evaluate? Plan? 如果是, 请说明影响和如何评估, 计划? (例如: 仿真分析等) Whether need sample to test? Plan? 是否需要样件进行验证? 计划?		Validation/签名 <u>徐润</u>
4. Product document influenced?		
4.1 Interface FMEA-relevant / IFMEA:	☐ no/否 ☐ yes/是 Note	
4.2. Product FMEA-relevant / DFMEA:	☐ no/否 ☐ yes/是 Note	
4.3. Special Charecteristics /PSC: (针对C Sample)	☐ no/否 ☐ yes/是 Note	
4.4. IMDS relevant: (针对C Sample)	☐ no/否 ☐ yes/是 Note	
4.5 Offer drawing relevant:(针对C Sample)	☐ no/否 ☐ yes/是 Note	
4.6. TCD relevant:	☐ no/否 ☐ yes/是 Note	Design /签名 <u>马坤阳</u>
5. Changed parts Stock& treatment / 需要变更零件的库存统计和处理? (只针对涉及到已经生产过的零件变更, 新放行零件和本条不相关)		☐ NA.(新零件放行,不涉及库存) ☐ 涉及到已经生产过的零件变更
RBCW RDC(外库)和RBCW 仓库, 是否有库存,	☐ 无库存 ☐ 有库存, 库存数量	
供应商处(已做好, 未发货)是否有库存,	☐ 无库存 ☐ 有库存, 库存数量	
LPN工厂是否有库存,	☐ 无库存 ☐ 有库存, 库存数量	
在途, 运输中是否有库存	☐ 无库存 ☐ 有库存, 库存数量	PUE /签名 <u>李奕岑</u>
被影响到的实物库存处理指示: (设计工程师组织会议讨论)	☒ 可以继续使用完 ☐ 改制后使用 ☐ 报废 ☐ 制费用; 报废成;	Design /签名 <u>马坤阳</u>
已装车件处理方法 (发客户或标定车等) (SMP组织会议讨论)	☐ 制; 更换; 其他 (单独说明);	PUE /签名
6. Influence on supplier part?		
Was this change confirmed with Purchasing (Supplier) for feasibility? Feedback from Supplier? 变更后的零件, 生产可行性供应商是否确认? 反馈?		
Influence on purchasing part delivery time? 对采购件交货时间的影响?		
涉及到变更的零件, 博世已出PR, 供应商还未来得及生产的订单, 是否已经通知供应商取消.		
Does the change influence on purchasing cost? 变更对采购成本的影响? ☐ no/否 ☐ yes/是		
☐ increase/增加, 金额 ☐ decrease/降低, 金额 ☐ no change/不变		PUE/签名 <u>李奕岑</u>
7. DFMA & Sample prodction OPL? 必要时附加DFMA 文件		
无OPL		Design /签名 <u>马坤阳</u>
8. Influence on LPN manufacturing&assembly process?		
Feedback from manufacturing engineer? 生产制造装配是否影响, 反馈?		
Influence on project&product delivery time? 对项目&产品交货时间的影响?		
Influence on manufacturing cost / 对生产成本的影响? ☐ increase/增加, 金额 ☐ decrease/降低, 金额 ☐ no change/不变		PUE/签名 <u>李奕岑</u>
9. PM alignment? (Influence on project Timeline? Cost?): 是否同意对项目时间和成本的影响; 需求样品订单数量		PM/签名 <u>张成</u>
10. 如果涉及以下情况需TCR3 审核确认。 不涉及则 NA.: ☐ 库存报废, 金额; ☐ 采购成本增加, 金额; ☐ 生成本增加, 金额		RBCW/TCR3
Design Team Leader	ENG-CV1	PUR
Signature / 签名: <u>杨研松</u>	Signature / 签名: <u>马坤阳</u>	Signature / 签名: <u>李奕岑</u>
		TCR3
		Signature / 签名: <u>李奕岑</u>

ATS RACE Workflow

ATS RACE release Checklist(c to D sample drawing release)

Release Points				
		Result/Status	Deadline	Resp.
Design	1. Final 3D model and offer drawing officially confirmed by customer?	} OK		2025.12.02
	2. Final nameplate confirmed by customer?			
	3. IMDS or CAMDS finished?			
	4. D-FMEA closed and signed ? Sign?	} covered by 493		
	5. I-FMEA signed with customer? Sign?			
	6. Installation instruction finished?			
	MOE	7. Final SC/CC finished by EGT5/MOE/QMM/PUR? Sign?	covered by 493	
1. Final welding process flow chart finished by MOE?		done		
2. Welding and assembly trial run finished?		} OK		2025.12.02
3. P-FMEA signed ?				
Validation	1. 9000 Km bad road Vehicle validation status?	} OK (see 493)		
	2. PV report?			
System	1. Customer validation report? (If customer take responsible the calibration, take as reference)	Covered by 9000 Km		2025.12.02
	2. TCD is available?	OK		2025.12.02
	3. PD is available?	OK		2025.12.02
Flow	1. QG0 and QG1 passed? QGC2 plan?	N/A		2025.12.02

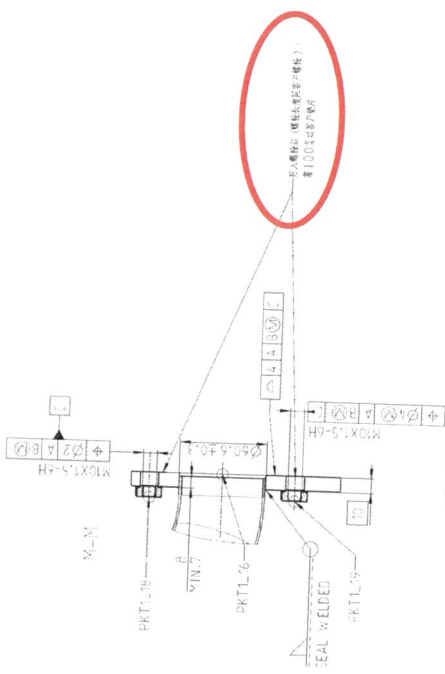
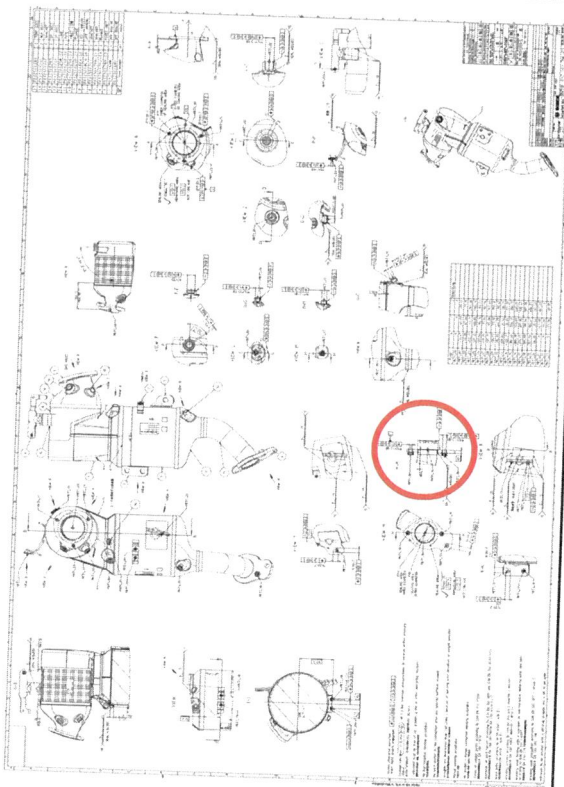
** :  Finished w/o open points; RACE can be proceeded.
 Finished w/ open points and countermeasures are available. Escalation to SM and inform to PM for RAC release or not.
 Finished w/ open points but root cause is not clear and countermeasure is not available; Inform PM and reject the RACE with inputting the reason in reject step.
 *** : Design SPJM responsible for the checklist document, but relative points should be checked with the Resp. person.



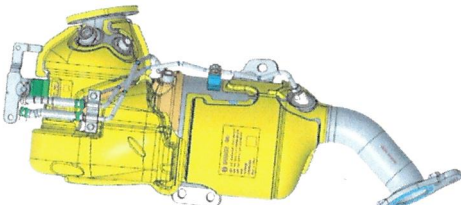
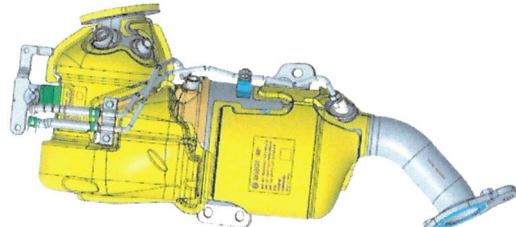
PDEC R25-128 (4JJ1-D 样件)

ENG-CV

PDECR 25-128

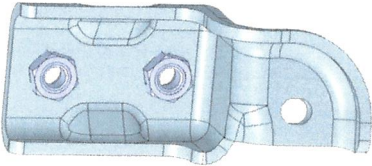
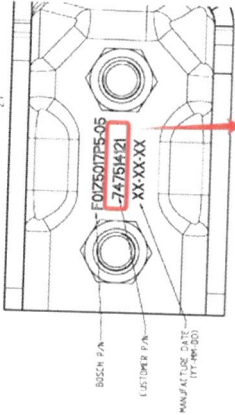
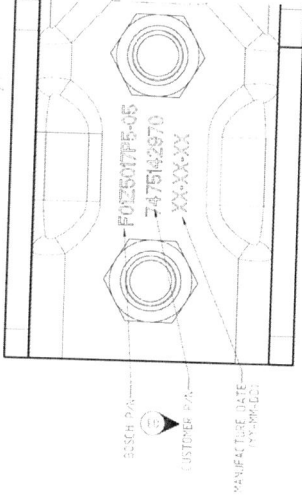
	零件号	零件名称	Project	样件阶段	变更来源
	F01Z5017NL-05	DOC+SDPF WELDING	JIE-4JJ	D	BOSCH/Customer
	变更前				
	F01Z5017NL-03				
变更描述	 增加新的检测方法: 拧入螺栓后 (螺栓长度同客户螺栓), 需100%过客户垫片				
变更原因	增加新的检测方法;				
风险点 (影响)	无				
验证方法	无				

PDECR 25-128

零件号		零件名称	Project	样件阶段	变更来源
F01ZH003EA-00		DOC+SDPF UNIT	JIE-4JJ	D	BOSCH/Customer
		变更前	变更后		
变更描述			 料号更改，模型无变更 由F01ZH003EA-03改为F01ZH003EA-00		
变更原因	料号更改				
风险点（影响）	无				
验证方法	无				
Internal Powertrain Solutions RBCW/ENG-ATS2 2023-09-22					



PDECR 25-128

	零件号	零件名称	Project	样件阶段	变更来源
	F01Z5017P5-05	BRACKET	J1e-4JJ	D	BOSCH/Customer
	变更前				
	F01Z5017P5-05			变更后	
变更描述					客户料号更新为 7475142970
变更原因	客户要求				
风险点 (影响)	无				
验证方法	无				



PDECR 25-128

零件号		零件名称	Project	样件阶段	变更来源
F01ZH003EB-00		SCR	Jle-4JJ	D	BOSCH/Customer
F01ZH003EB-05		变更前	变更后		
变更描述		 QR NAME PLATE 二维码位置:置于密封螺帽内部, 盖帽盖紧时在壳壁上 材料: CR 内容除标签数据一致 CUSTOMER P/N SUPPLIER CODE BOSCH P/N MANUFACTURE DATE (YYYYMMDD) SERIAL NUMBER 生产日期和序列号需满足参数 CA100274701*E415 *747514122*F01ZH 003EB*XXXXXXXX XXXXXX (24) (24) (60) (80)  CA100274700*E415 *747514122*F01ZH 003EB*20251021 00006 此次图纸更新件号升级为 CA100274701			
变更原因		SCR 客户要求 在尾管处增加纸质标签			
风险点 (影响)		无			
验证方法		无			
Internal Powertrain Solutions RBCW/ENG-ATS2 2023-09-22					
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PDECR 25-128

零件号		零件名称		变更来源	
F01Z5102T9-00		DPF MAT		BOSCH/Customer	
变更前		变更后			
Project		样品阶段			
Jle-4JJ		C			
变更原因		工厂反馈衬垫封装后，会超出催化剂，建议剪短5mm。经奇耐确认剪短5mm后，安全系数为1.1，评估风险较低，可以剪短			
风险点 (影响)		无			
验证方法		无			

变更描述

内部展示 Special Requirements:

客户确认日期: 2023.09.22

Dimension/尺寸	A[mm]	B[mm]	C[mm]	D[mm]	E[mm]	F[mm]	Thickness[mm]	Base Weight[kg/m2]	Part Weight[kg]
Nominal/标称	578.0	163.0	40.0	53.0	55.0	618	13.50	1800	169.59
Tolerance/公差	±2.5	±1	±1	±1	±1				
Max.重量	580.5	164.0	41.0	54.0	56.0		14.9	1844	186.54
Min.最小值	575.5	162.0	39.0	52.0	54.0		12.1	1656	152.63

制图/日期: 客户确认日期

备注/Remark:

内部展示 Special Requirements:

客户确认日期: 2023.09.22

Dimension/尺寸	A[mm]	B[mm]	C[mm]	D[mm]	E[mm]	F[mm]	Thickness[mm]	Base Weight[kg/m2]	Part Weight[kg]
Nominal/标称	578.0	158.0	40.0	53.0	55.0	618	13.50	1800	164.35
Tolerance/公差	±2.5	±1	±1	±1	±1				
Max.重量	580.5	159.0	41.0	54.0	56.0		14.9	1844	169.32
Min.最小值	575.5	157.0	39.0	52.0	54.0		12.1	1656	161.64

制图/日期: 客户确认日期

备注/Remark:



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Distribution: RBCW/TCR3-Xie Youfu, /ENG-CV1-Jiang Xingkai, /ENG-ATS-Luo Jin,
/ENG-ATS-Yang Guangtuo, /SMP-ATS1-Zhang Shengchang

Page 1 of 11

Tere25_029

Test Report

Vehicle endurance test parts investigation

1. General information

Project name	JIE 4JJ ATS CN6	MCR No.	-
ATS concept	☒ in-line ☐ Box ☐ U-shape	Test ground	XIANGYANG
Part number	DOC+DPF: F01ZH003EA-02 SCR: F01ZH003EB-02	ATS sample phase	B sample

2. Operation condition

Test item	Comprehensive roads endurance test
Test description	As the design validation requirement for JIE 4JJ ATS CN6 project, 42000km comprehensive roads endurance test was performed to the ATS parts. The test purpose is to verify the resistance to the rough road load for ATS mechanical parts.

3. Test result

- 1) Sealing check
The leakage rate of the ATS after test (DOC+DPF:0.157L/min, SCR:0.035L/min) can fulfill the design requirement (DOC+DPF:<0.5L/min, SCR:<0.3L/min @filling pressure 30 ± 5 kpa).
- 2) Visual inspection
All components visual check are in acceptable condition.

4. Conclusion

☒ Pass ☐ Pass with limitation ☐ No pass



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Page 2 of 11

Tere25_029

5. Investigation

1) Leakage test for the whole ATS

- After the endurance test, the leakage test was performed to the DOC+DPF and SCR units. From the leakage test results (table 1-2), the leakage rate of the ATS after test can fulfill the design requirement.

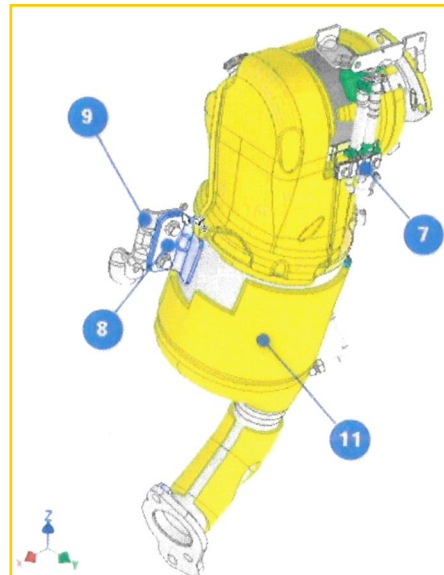
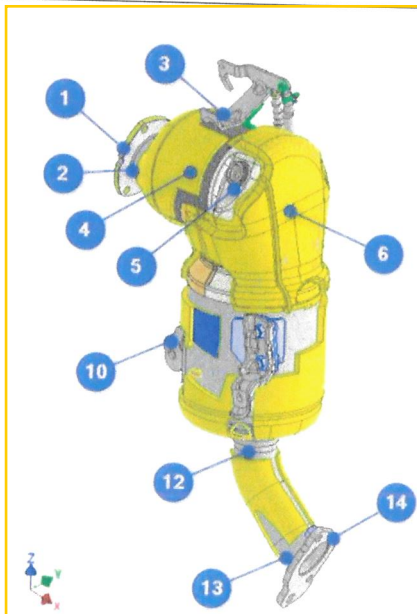
Table 1 Leakage test result for DOC+DPF

Leakage rate result before and after test @30kPa filling pressure		
Leakage rate (L/min)	After testing	Test requirement
	0.157	<0.5

Table 2 Leakage test result for SCR

Leakage rate result before and after test @30kPa filling pressure		
Leakage rate (L/min)	After testing	Test requirement
	0.035	<0.3

2) DOC+DPF investigation after test



Part No.	components	No failure	Failure	Figure
1	DOC inlet flange	×		Fig.3
2	DOC inlet flange welding seam	×		Fig.4-5
3	ΔP sensor bracket and welding seams	×		Fig.6-7
4	DOC catalyst	×		Fig.9-10



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Page 3 of 11

Tere25_029

5	DM flange	x		Fig.21-22
6	Mixer plate	x		Fig.11-12
7	Δ P pipe bracket and welding seams	x		Fig.8
8	DPF left mounting bracket 1	x		Fig.13-14
9	DPF left mounting bracket 2	-		-
10	DPF right mounting bracket	x		Fig.15-16
11	DPF catalyst	x		Fig.17
12	DOC outlet flange welding seam	x		Fig.18-19
13	DOC outlet flange	x		Fig.20

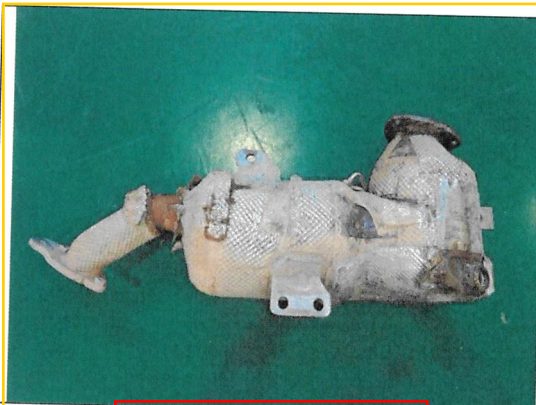


Fig.1 DOC+DPF overview



Fig.2 DOC+DPF overview

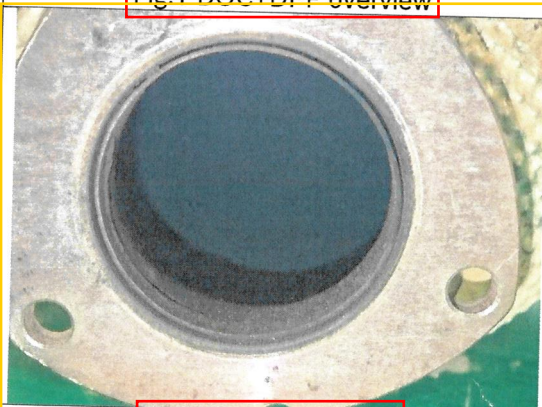


Fig.3 DOC inlet flange



Fig.4 DOC inlet flange welding seam



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Page 4 of 11

Tere25_029



Fig.5 DOC inlet flange welding seam



Fig.6 ΔP sensor bracket and welding seams

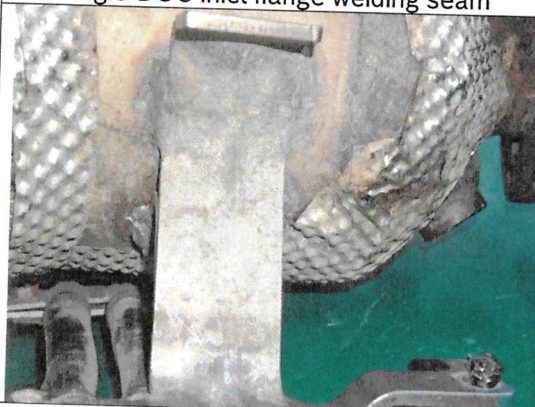


Fig.7 ΔP sensor bracket and welding seams

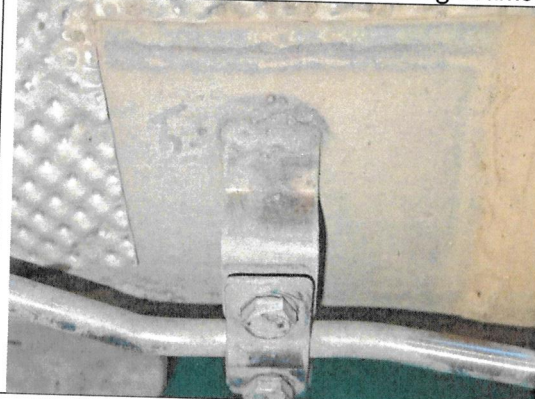


Fig.8 ΔP pipe bracket and welding seams

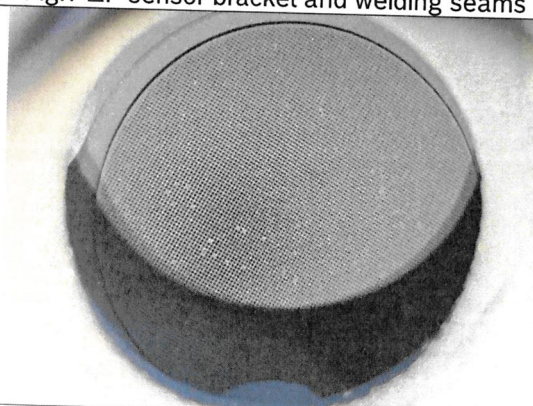


Fig.9 DOC catalyst

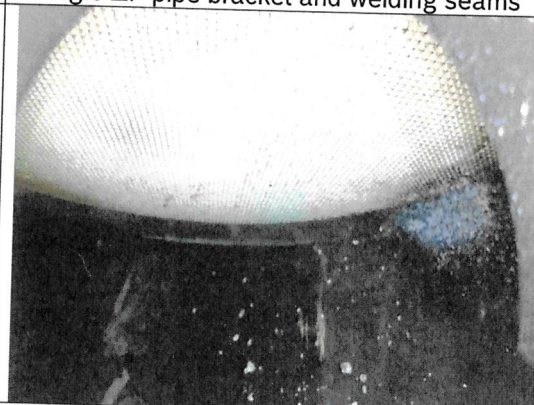


Fig.10 DOC catalyst



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Page 5 of 11
Tere25_029



Fig.11 Mixer plate

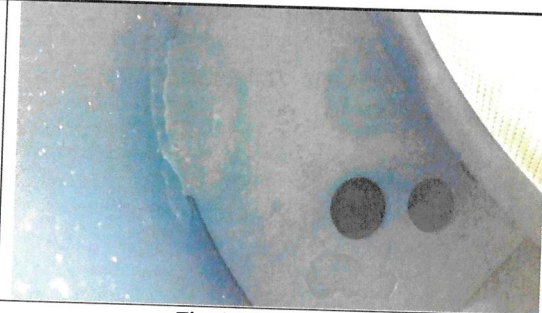


Fig.12 Mixer plate



Fig.13 DPF left mounting bracket 1



Fig.14 DPF left mounting bracket 1

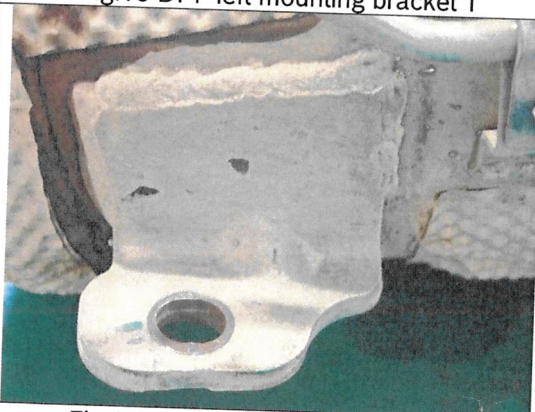


Fig.15 DPF right mounting bracket

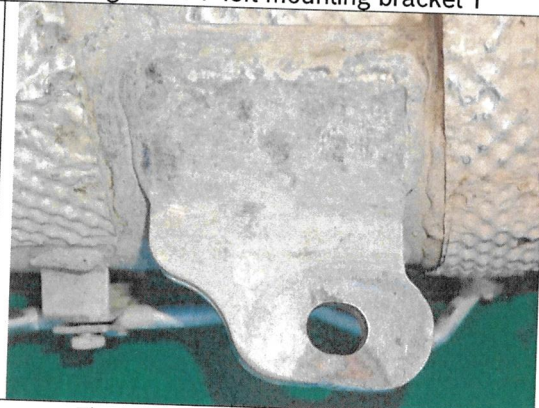


Fig.16 DPF right mounting bracket

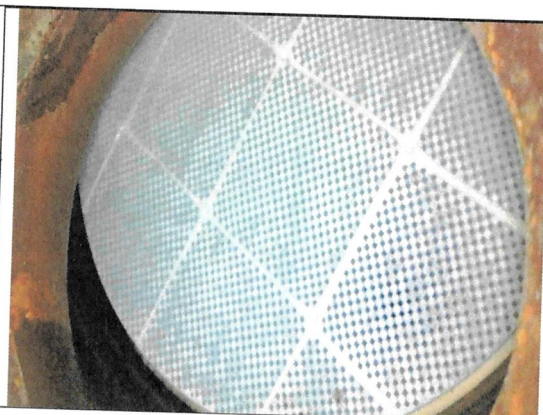


Fig.17 DPF catalyst



Fig.18 DPF outlet flange welding seam



Fig.19 DPF outlet flange welding seam



Fig.20 DPF outlet flange welding seam



Fig.21 DM flange and welding seam



Fig.22 DM flange and welding seam



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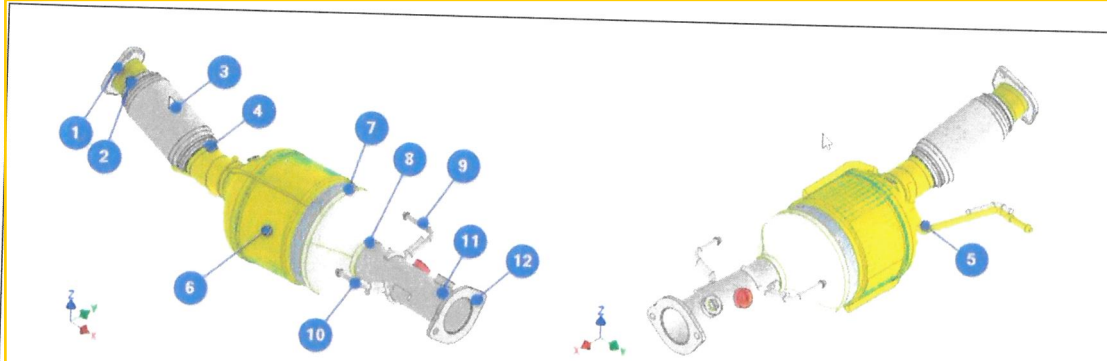
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Page 7 of 11

Tere25_029

3) SCR investigation after test



Part No.	components	No failure	Failure	Figure
1	SCR inlet flange and welding seams	×		Fig.25-27
2	Flexible inlet welding seam	×		Fig.28-29
3	Flexible pipe	×		Fig.28-29
4	Flexible pipe outlet welding seam	×		Fig.28-29
5	Front hook and welding seam	×		Fig.30-32
6	SCR catalyst	×		Fig.33-34
7	SCR sleeve welding seam	×		Fig.23-24
8	SCR outlet pipe welding seam	×		Fig.35-36
9	Right rear hook and welding seam	×		Fig.37-38
10	Left rear hook and welding seam	×		Fig.39-40
11	SCR outlet flange welding seam	×		Fig.41-42
12	SCR outlet flange	×		Fig.43



Fig.23 SCR overview



Fig.24 SCR inlet flange



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Page 8 of 11

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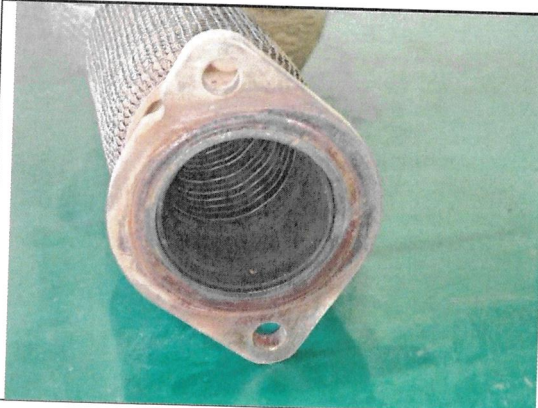


Fig.25 SCR inlet flange



Fig.26 SCR inlet flange welding seam

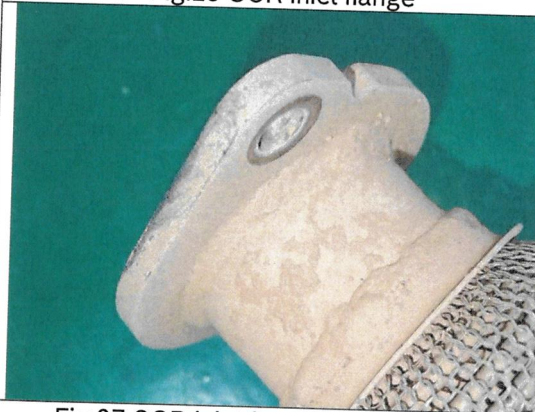


Fig.27 SCR inlet flange welding seam



Fig.28 Flexible pipe outlet welding seam



Fig.29 Flexible pipe outlet welding seam



Fig.30 Front hook and welding seam



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Page 9 of 11

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Fig.31 Front hook and welding seam



Fig.32 Front hook and welding seam



Fig.33 SCR catalyst

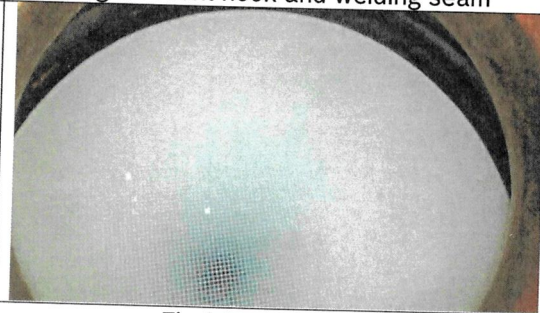


Fig.34 SCR catalyst



Fig.35 SCR outlet pipe welding seam

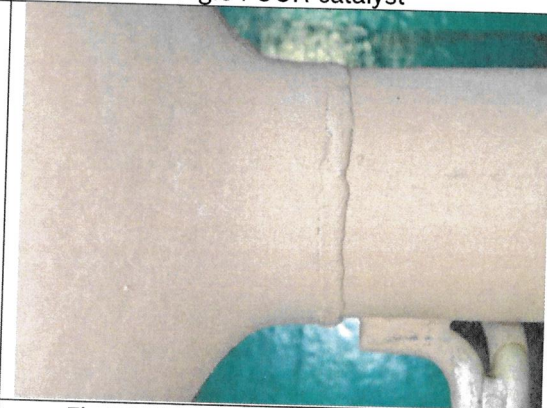


Fig.36 SCR outlet pipe welding seam



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Page 10 of 11

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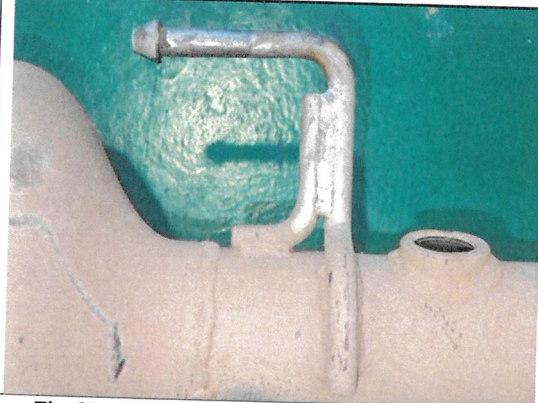


Fig.37 Right rear hook and welding seam



Fig.38 Right rear hook and welding seam

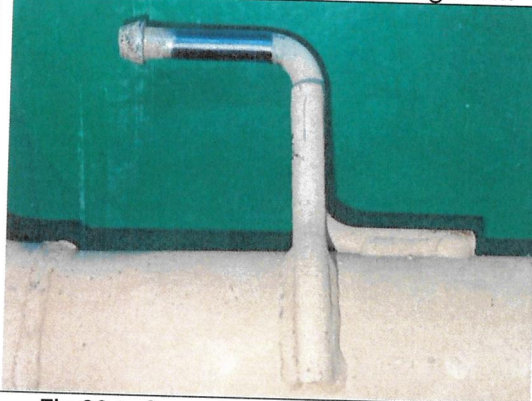


Fig.39 Left rear hook and welding seam



Fig.40 Left rear hook and welding seam



Fig.41 SCR outlet flange welding seam



Fig.42 SCR outlet flange welding seam



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Page 11 of 11

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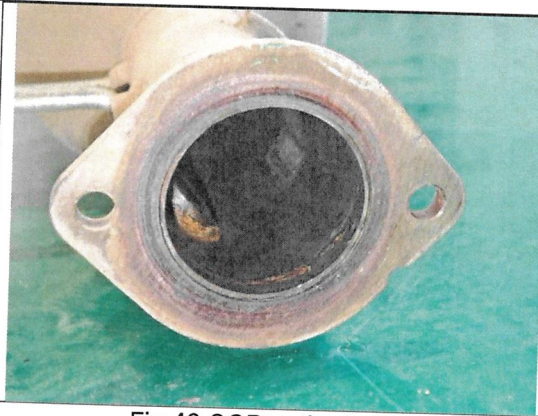


Fig.43 SCR outlet flange